

Editor comments:

Authors are thanked for their revised manuscript. While the reviewer was satisfied with the changes made, the manuscript still has a number of outstanding issues and requires further revision to bring the work up to a publishable standard. Line numbers below refer to the clean version of the manuscript

1. The standard of language in the manuscript still falls well short of that required for publication. The issues are minor but extensive. Authors must seek professional language assistance. There are spelling issues, grammatical issues, non-scientific language, incorrect tenses, and awkward phrasing throughout.

Authors' response: We appreciate your feedback on this matter. Following your suggestions, we have sought language assistance to improve the quality of the manuscript. We sincerely hope that the revised manuscript is up to standard in terms of language.

2. Highlights are not adequate, suffering from language issues and imprecise phrasing.

Authors' response: Thank you for raising this issue to our attention. We have revised and corrected the highlights.

3. L16: "Histological monitoring showed recruitment of amoebocytes, the immune effector cells in anthozoans, and changes to mesoglea organization between 6 h and 24 h post-injury that revert by 7 days." Authors generally do a good job throughout the manuscript in making sure that they do not infer quantitative changes from their histology data. However, this sentence does make the kinds of comparisons that cannot be justified with a statistical analysis, and so will require editing.

Authors' response: Thank you for pointing out this issue. We have revised and rewritten the sentence as follows: "*Histological monitoring showed signs of amoebocyte recruitment, the immune effector cells in anthozoans, and fibrillar mesoglea organization detectable between 6 h and 24 h post-injury*".

4. L53: Lipid peroxidation is not limited to PUFA's though

Authors' response: Thank you for your comment. We have changed the sentence into "*While ROS signaling and production perform important roles during these processes, imbalances between ROS generation and scavenging can result in damage to cellular lipids, proteins or DNA*".

5. L59 and throughout: do not italicise chemical formulae.

Authors' response: Thank you for your comment. Italics have been removed from all chemical formulae in the manuscript.

6. L59: "H₂O₂ is less dangerous" Unless this can be supported with a literature-based argument it seems speculative, and should be edited.

Authors' response: Thank you for your suggestion. The statement has been rewritten and further referenced to improve accuracy:

"Because H₂O₂ is not charged, it can diffuse across biological membranes and cause oxidative damage not restricted to the generation site (Lesser, 2006). H₂O₂ scavenging is crucial to avoid generation of the more reactive hydroxyl radical, and both catalase (CAT) and glutathione peroxidase (GPx) fulfill this role in different cellular compartments (Lesser, 2006; Lushchak, 2011)".

7. L118: "water temperature varied with ambient temperature," Please give the temperature range.

Authors' response: Thank you for your comment. We have detailed maximum and minimum temperatures in L123 of the revised manuscript.

8. L121: Please provide more details regarding the sectioning procedure. How was this performed?

Authors' response: We appreciate your question. We have included more details on the sectioning procedure in L125 of the revised manuscript.

9. L131: Presumably, tentacles and columns were dissected prior to sampling and freezing? If so, then please reorder the text here so that it occurs in a logical sequence.

Authors' response: We appreciate your question. In this specific case, dissection of the tentacles and columns was carried out after freezing the individuals, before homogenizing.

10. L171: State the two pH values used. Indicate how pH's were adjusted.

Authors' response: We appreciate your suggestion. We have added pH values used and more details about how they were adjusted. The sentence has been rewritten as follows:

"Acid (EC 3.1.3.2) and alkaline (EC 3.1.3.1) phosphatase (ACP and ALP respectively) activities were measured following the method of Huang et al. (2011), based on the change of absorbance produced by the activity of the enzyme on p-nitrophenyl at pH 5.0 and 8.0, respectively. Solution pH was adjusted through addition of HCl or NaOH."

11. L180: Again, please give temperature range.

Authors' response: We appreciate your recommendation. We have detailed maximum and minimum temperature for the Histology subsection in L188 of the revised manuscript.

12. L183: Please provide size.

Authors' response: Thank you for this remark. The size of the anemones has been explicitly stated in L190 of the revised manuscript.

13. L207: "Consistently with previous observations," This is interpretive. Results should be descriptive only.

Authors' response: Thank you for pointing this out. This interpretive statement has been removed.

14. L209: For clarity, authors should state that the mortality data are not shown (i.e., insert "(data not shown)" at the end of this sentence.

Authors' response: Thank you for pointing this out. We have explicitly stated that the data is not shown, in L217 of the revised manuscript..

15. The term "section" is a bit confusing throughout the Results. At initial reading I had assumed that section referred to column and tentacles as the two different sections of the animal. The term "treatment" would be better, as that is more obvious and representative.

Authors' response: Thank you for your comment on this matter. Following your suggestion and to reduce confusion, we have changed the term "section" to "treatment" in the result section text and figures. Mentions of "sectioned anemones" have been changed to "bisected anemones" throughout the text. We hope these changes will improve clarity of the manuscript.

16. The use of estimated marginal means is fine, but there are some issues here. First, the two way ANOVA approach allows authors to account for the effects of time in determining if there was an effect of treatment. This is good and important. However, ultimately, whether the basal values of the endpoints varies with time is not really pertinent to the hypotheses being tested. Therefore, the estimated marginal means approach should really only be applied to data where there is a significant effect of treatment, and a significant interaction effect. In the first two figures this only applies to the tentacle NOQ1. With this in mind, then the "emm" figure should move from Figure 4 and become Figure 2C. Please note that this also solves the problem that figures must be numbered in the order of their appearance in the text, which currently does not occur in this submission. Then there is the issue that there are no statistical indicators in any of the other figures, which means that the figures are very difficult to interpret. For the tentacle NOQ1 authors can indicate with an asterisk to see Figure 2C, but in all other Figure 1 and 2 panels, it is suggested that authors use an asterisk to indicate effects of treatment. In the figure legend this approach can be explained (i.e., effects of time not indicated, when there is an effect of treatment but significant interaction see the emm panel).

Authors' response: Thank you for expressing your concerns regarding this matter. We completely agree on the order of Figures, and as such we have split Figure 4 (displaying estimated marginal means and confidence intervals) into three figures so that they match the order of mention in the text. We have also added asterisks as

statistical indicators in the rest of the boxplots showcasing a significant treatment main effect, and updated the captions accordingly.

However, while it is true that differences solely due to time are not the focus of our research questions, the presence of a significant interaction effect indicates that effects of the two variables cannot be evaluated independently, whether the main effects are significant or not.

In cases where there is no significant main effect of treatment, but there is one for time and interaction, it may be the case that there are still differences associated with treatment occurring at only one level of the variable time. These differences are indeed pertinent to our hypothesis, as there might be an effect at T1 but not at T2. In our manuscript, this is the case in tentacular GPx, tentacular GST and columnar MPO. Moreover, even if the main effects are not significant, a significant interaction can still occur (i.e.: tentacular catalase in our manuscript) if the effect of the first variable (treatment) has an opposite direction at each level of the second variable (time). For all of these reasons we consider that evaluating interaction effects is valuable and relevant even in those cases where the two-way anova did not result in a significant treatment main effect.

17. L228: “However, tentacular GST activity was significantly affected by section ($F(1,32)= 8.76$, $p = 0.030$), time ($F(1,32)= 23.91$, $p = 0.001$) and interaction ($F(1,32)= 17.58$, $p = 0.003$).” This is an error. These data refer to NOQ1 in tentacle and not tentacle GST.

Authors’ response: Thank you for raising this issue to our attention. We have corrected the mistaken reporting of results in L237, which now reads as follows: *“However, tentacular GST activity was significantly affected by time ($F(1,32) = 23.91$, $p = 0.0008$) and interaction ($F(1,32) = 17.58$, $p = 0.004$).”*

18. There is still an issue with the TEAC units. How do you have a volume (micromoles per litre) for a tissue measurement? If the volume here refers to a supernatant then this value will reflect differences in tissue mass and buffer volumes and not actual activity. These data must be expressed per unit protein, which normalizes data to the actual amount of tissue.

Authors’ response: Thank you for reiterating the need for this modification. We have expressed TEAC in $\mu\text{mol Trolox equivalent} / \text{mg of tissue}$.

19. Table 1 should be moved to supplemental material, and all abbreviations must be defined in the table footnote.

Authors’ response: Thank you for your recommendation. We have moved Table 1 to Supplementary Materials and have included definitions of all the abbreviations in the table caption.

20. L279: “appeared to be laxer” Cannot state without statistical analysis.

Authors' response: Thank you for your comment. We have eliminated the comparative statement.

21. Figure 7 caption: Again "section area" is confusing. This is a tissue section, but authors are referring to the dissection. Please also add the definition of the yellow line to Figure 9 caption.

Authors' response: We appreciate your suggestion. We have changed "section area" to "bisection plane". Description of the yellow line has been added to Figure 9.

22. L294: Any paragraph that does not refer to the study outcomes is not a Discussion, but an Introduction. Integrate this paragraph with the following paragraph to make a more cohesive discussion.

Authors' response: Thank you for your recommendation. Following your suggestion, we have moved some ideas presented in the paragraph to the Introduction section (L52 & L59). The remaining ideas of the paragraph have been integrated with the following paragraph in L303.

23. L301: 'The affected enzymes were mainly glutathione dependent enzymes (such as GPx, GR and GST) and/or H₂O₂-processing enzymes (CAT and GPx).' This is not true. The only significant effects of treatment were on column GR and column CAT.

Authors' response: We appreciate your comment on the matter. As previously mentioned, a non-significant main effect of treatment does not rule out the possibility of the dependent variable being affected by the interaction of treatment and time, meaning that the effect of treatment would be dependent on the time point at which the measure was taken. This scenario is relevant for our research question, and applies to tentacular CAT, GPx and GST, where mean activity at T1 was higher in bisected anemones than controls, despite this trend disappearing or reversing at T2.

To clarify the rationale behind this, we've modified the previous sentence to read as follows: "*The present study detected increases in columnar and tentacular antioxidant enzyme activities in regenerating anemones **at one or both** time points [...].*"

24. L309: 'The increased activity of this enzyme on sectioned anemones could be related to the increased aerobic metabolic rate that is expected of animals during regeneration processes (Mack and Bely, 2025).' This generalization needs to be explained. Be explicit with the rationale for this statement.

Authors' response: We appreciate your recommendation. We have explicitly stated the reasoning behind this sentence, expanding it as follows:

"The increased activity of this enzyme on bisected anemones could be the consequence of a greater generation of endogenous quinones, as these have been found to induce NQO1 expression in mammals (Atia and Abdullah, 2020). NQO1 also has superoxide scavenging activity, and has been reported to be over-expressed in mammalian tissues where SOD expression is low (Dinkova-Kostova et al., 2004;

Siegel et al., 2004). Ultimately, a higher activity of NQO1 could offer an extra layer of antioxidant protection against the oxidative stress associated with the metabolic activation known to occur during regeneration processes (Atia and Abdullah, 2020; Mack and Bely, 2025)."

25. L318: "featured slightly lower levels" Please be accurate. This was "significantly lower". "Slightly" is ambiguous.

Authors' response: Thank you for raising this to our attention. We have removed the word "slightly" from the expression, which now reads "*significantly lower lipid peroxidation*".

26. L356: "because ACP and ALP levels were still abnormally high at 20-weeks post injury." What is the basis of this suggestion? How do these activities compare to the literature? Please also be accurate and refer to enzyme measurements as "activities" not "levels"

Authors' response: We appreciate your question. Mentions of "levels" have been replaced by "activities" We have reconsidered the reasoning in the paragraph and have rewritten the ideas in it more clearly, as follows:

"In contrast with the evident antioxidant response, we detected no differences in ACP and ALP activity between control or sectioned anemones throughout the study. Parisi et al. (2021) reported increases to ALP activity in A. viridis after tentacle amputation extending to 14 days post-injury. This discrepancy with our results suggests that the enzymatic inflammatory response occurring after an injury is not detectable after 4 weeks post-injury. Histological examination supports this idea, as it confirmed that wound healing occurred within the first week after the injury in the case of longitudinal sections (Figure 7).